

Dynamic No-Trade Bands for Intraday Delta Hedging

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July 2026

1 Model Framework

1.1 From a Static Band to a State-Dependent Band

The static strategy holds a constant no-trade band H around $\Delta_{\text{target}} = 0$ and hedges only when $|\Delta(t)|$ clears it. A constant H encodes a constant answer to the question the band actually poses: *how much residual delta is worth carrying rather than paying to remove?* That answer is not constant. The cost of carrying delta over the next interval depends on how fast the book regenerates delta once hedged, and that speed is a measurable state variable.

The dynamic model therefore replaces the constant with a per-bar band:

$$-H(t) \leq \Delta(t) \leq H(t), \quad |\Delta(t)| > H(t)(1 + \epsilon) \Rightarrow \text{hedge.} \quad (1)$$

Everything else in the strategy — execution scheduling, cooldown, cost and risk measurement, benchmarks — is unchanged. The static rule is the special case $H(t) \equiv H_0$, so the dynamic model strictly nests it and any improvement must be earned.

1.2 The Gamma-Conditioned Band

Between trade arrivals the book's delta does not sit still: it drifts with the underlying at a rate set by the book's convexity, $d\Delta \approx \Gamma dS$. Gamma is thus the *speed* at which market moves regenerate delta after a hedge. For a band of half-width H and per-bar underlying moves of scale σ_S , the expected number of bars for gamma drift alone to carry delta across the band is of order

$$\left(\frac{H}{|\Gamma| \sigma_S} \right)^2.$$

A high-gamma book traverses a fixed band quickly and therefore carries more unhedged risk between checks; a low-gamma book barely moves and the same band is needlessly tight. Making the band shrink as gamma rises equalizes this breach behaviour across regimes:

$$H(t) = \frac{H_0}{1 + c|\Gamma(t)|}. \quad (2)$$

- H_0 : base band half-width, in hedge-instrument units — the band the rule uses when the book is gamma-flat;
- $\Gamma(t)$: book gamma at bar t , in hedge-instrument units per unit move of the underlying;
- c : tilt strength, controlling how hard the band responds to gamma.

Units of c . Gamma enters undivided, so c is *not* dimensionless: it carries the reciprocal units of Γ , and the band halves exactly when $|\Gamma| = 1/c$. The useful range of c is therefore set by the book’s own gamma scale and differs by orders of magnitude across instruments (Table 1). Dividing $|\Gamma|$ by its sample median and rescaling c gives the identical family of bands with a dimensionless c ; only the units of the parameter change, not the model.

Limiting cases. $c \rightarrow 0$ recovers the static band $H(t) \equiv H_0$; large c drives $H(t) \rightarrow 0$ whenever the book carries gamma, approaching continuous reheding. The backtest selects c , so the data decides how hard to tilt.

The sign of the effect is not obvious. Classical optimal-hedging asymptotics under proportional transaction costs (Whalley–Wilmott; Zakamouline) give an optimal half-width $\propto \Gamma^{2/3}$ — i.e. *wider* when gamma is high, on the grounds that chasing a fast-moving delta churns execution costs. Equation (2) tilts the other way, prioritising risk control over turnover. Which effect dominates depends on the cost model and the risk measure, so c is gridded and $c \rightarrow 0$ recovers the static band: the comparison is decided empirically, not assumed.

1.3 Book Gamma Revaluation

The band in Equation (2) requires $\Gamma(t)$ at every bar. It is built exactly like the book delta, by repricing every live position at the current underlying price and aggregating:

$$\Gamma(t_j) = \sum_{i: t_i \leq t_j} s_i n_i \gamma_i(t_j) m, \quad \gamma_i(t_j) = \frac{\varphi(d_{1,i}(t_j))}{S_j \sigma \sqrt{\tau_i(t_j)}}. \quad (3)$$

- $\gamma_i(t_j)$: per-contract gamma of option i at bar t_j (identical for calls and puts);
- $\varphi(\cdot)$: standard normal density; $d_{1,i}$ is as in the delta definition;
- S_j : underlying price at bar t_j (futures price F_j for ES);
- m : contract-to-hedge-unit multiplier ($m = 1$ for ES options, whose deltas are already in futures equivalents; $m = 100$ for equity options, which deliver 100 shares);

- s_i, n_i, σ, τ_i : as in the delta definition — trade sign, contract count, the day’s flat volatility input, and time to expiry.

Gamma is strategy-independent. The hedge instrument — a future or the stock itself — has zero gamma, so hedging changes Δ but never Γ . The path $\Gamma(t)$ therefore depends only on the option tape and the price path, not on the hedging rule, and is precomputed once per day and reused by every configuration in the grid. This is what makes the dynamic grid search no more expensive than the static one.

1.4 Instrument Instantiation

The framework above is instrument-agnostic; only the units and the gamma scale change.

	ES futures	SPCX equity
Hedge instrument	ES future	The stock
Delta / band unit	Contract equivalents	Shares
Contract multiplier m	1	100
P&L multiplier M_h	50 \$/point	1 \$/share
Option delta model	Black-76 (futures)	Black-Scholes (spot)
Γ units	contracts/point	shares/\$
Median $ \Gamma $	0.0928	2,176
c units	points/contract	\$/share
c grid	$\{5, 10, 20\}$	$\{2, 5, 10, 20\} \times 10^{-4}$

Table 1: Instantiation of the dynamic band model on the two books. The c grids are *not* interchangeable: each is scaled to its own book’s gamma so that $c|\Gamma|$ spans roughly 0.5–2.

2 Strategy and Experimental Design

This section states the experimental protocol shared by both books. Only the parameter *values* in Table 2 differ between them.

2.1 Trigger Rule and Hedge Sizing

The band is monitored on a 5-minute grid. A hedge fires when Equation (1) is satisfied, and the parent order returns the book to flat:

$$\begin{aligned}
 Q_t &= -\Delta(t) && \text{(full flatten),} \\
 Q_t &= -\text{sign}(\Delta(t)) (|\Delta(t)| - H(t)) && \text{(edge target).}
 \end{aligned} \tag{4}$$

The edge-target variant trades only the excess over the band and leaves the residual parked at the edge; it is evaluated as a refinement in both results

sections. After a schedule completes the trigger is suppressed for T_{cooldown} unless the breach is large ($|\Delta(t)| > 2H(t)$, with the override threshold scaling with the *current* band). Any residual delta is forcibly flattened at the close, so every configuration satisfies $\Delta(T) = 0$.

2.2 Execution

The parent order is split into child orders over a horizon τ , i.e. over $N = \tau/5$ min consecutive bars, either equally weighted (TWAP) or weighted by each bar's share of traded volume (VWAP). A new trigger cancels and replaces any running schedule, resized to the current residual delta.

2.3 Cost and Risk Measurement

Execution cost is a market-data proxy combining a half-spread term and a concave impact term:

$$C_{\text{exec}} = \sum_k M_h |q_k| \left[\frac{s}{2} + \alpha S_k \left(\frac{|q_k|}{L_k} \right)^\beta \right], \quad \beta = 0.5. \quad (5)$$

- q_k : signed child order in bar k ; S_k the underlying price there;
- $s/2$: half bid–ask spread in price units;
- α : impact level; L_k the liquidity the fill leans on.

α and L_k are the one place the two books genuinely diverge, and the divergence is deliberate rather than a porting artifact. On ES, α is anchored so a 1-lot pays ≈ 0.5 bp of notional at median ADV, and L_k is full-day ADV — both harmless for an instrument whose hedge clips are $\sim 10^{-4}$ of ADV. Neither survives on SPCX, where a hedge clip is a meaningful fraction of a five-minute execution window's liquidity even though it is small against full-day ADV: there $\alpha = Y\sigma_{\text{daily}}$ follows the square-root law of impact (Toth–Bouchaud, $Y = 0.5$), anchoring impact to the book's own volatility, and L_k is the liquidity of the fill's *execution window* ($\text{ADV} \times 5/390$ for an intraday child) rather than the whole day. Both books treat the final bar as an auction cross: zero spread, full-day depth, and an impact haircut.

Delta risk is the root mean square of realized residual-delta P&L per bar:

$$\ell_{j+1}^\Delta = M_h \Delta(t_j) (S_{j+1} - S_j), \quad C_{\text{risk}} = \sqrt{\frac{1}{M} \sum_{j=0}^{M-1} (\ell_{j+1}^\Delta)^2}. \quad (6)$$

2.4 Benchmarks and Standardized Objective

Zero Hedge carries the book all day and flattens once into the close (cost floor, risk ceiling); **Full Hedge** re-flattens every bar (risk floor, cost ceiling). Both bound the attainable region and normalize the objective:

$$L_{\text{std}}(\theta) = \frac{C_{\text{exec}}(\theta)}{C_{\text{exec}}(\text{Full Hedge})} + \lambda_{\text{risk}} \frac{C_{\text{risk}}(\theta)}{C_{\text{risk}}(\text{Zero Hedge})}, \quad \lambda_{\text{risk}} = 1. \quad (7)$$

2.5 Fair-Comparison Methodology

Two protocol choices carry most of the inferential weight, because the naive comparison is misleading.

Width matching. At the same H_0 the gamma band is *narrower on average* than the static band, since the divisor in Equation (2) is ≥ 1 . A comparison at matched H_0 therefore conflates two different things: the value of *conditioning the band on gamma* and the value of simply running a *tighter* band. Every headline dynamic-vs-static claim below is instead made against a static band set to the gamma configuration’s *mean effective width*, under identical ϵ , cooldown, horizon and method.

Hedge-frequency capping. The cost proxy in Equation (5) contains no floor on order splitting, and impact is concave in clip size ($Q^{1.5}n^{-1/2}$ for n children), so splitting a hedge into more, smaller child orders is strictly cheaper in the model without limit. When impact is large enough, unconstrained grid search runs away toward the tightest band available — a boundary solution, not a strategy — and capping the average number of hedges per day restores an interior optimum. Whether that device is needed is book-specific. On ES a fixed per-ticket cost bites (execution cost there is small enough that a plausible per-order charge moves the optimum). On the SPCX book the optimum is already an interior, wide band, so neither a per-ticket cost nor a frequency cap is required. The results sections quantify this for each book.

2.6 Parameter Grid

Parameter	Role	ES grid	SPCX grid
H_0	Band width	{2, 5, 10, 25, 50, 100, 200} ct	{5, 10, 25, 50, 100, 200, 300}k sh
c	Gamma tilt	{5, 10, 20}	{2, 5, 10, 20} $\times 10^{-4}$
τ	Horizon	{15, 30, 60} min	
e	Method	{TWAP, VWAP}	
ϵ	Buffer	5% (held fixed)	
T_{cooldown}	Cooldown	15 min (held fixed)	
λ_{risk}	Risk weight	{0.25, 0.5, 1, 2, 4} (sweep)	

Table 2: Strategy grid. ϵ and T_{cooldown} are held fixed across all configurations so that the static-versus-dynamic comparison is not confounded by the second-order trigger refinements.

Each configuration is simulated over every day in the sample using the same precomputed delta and gamma paths, so differences across configurations reflect the hedging rule alone.

3 Results: ES Futures

Data. 102 trading days (January 2 – May 29, 2026), a simulated tape of 32,127 executed ES option trades, and 8,364 five-minute futures bars. Cost model: $\alpha = 0.055$, half-spread 0.125 points, closing-auction impact factor 0.15, $M_h = 50$.

Benchmark	Exec. cost	Risk (RMS/bar)	L_{std}
Zero Hedge	\$235,240	\$24,801	1.058
Full Hedge	\$4,030,988	\$0	1.000

Table 3: ES benchmark bounds over 102 days.

Gamma signal. Book gamma has a median of 0.0928 contracts/point (90th/99th percentiles 0.368 / 0.999). Within a day $|\Gamma|$ has a coefficient of variation of 0.61 and a max/median ratio of 2.1 $\times$; across bars it correlates with $|\Delta|$ at $\rho = 0.63$. Gamma is thus a strongly varying but only partially independent signal on this book.

Family	H_0	c	Exec. cost	Risk	L_{std}
Static	2	—	\$974,803	\$6,270	0.4946
Combined	10	20	\$653,688	\$8,309	0.4972
Gamma	10	20	\$668,157	\$8,320	0.5012
Time-decay	10	—	\$721,897	\$8,331	0.5150

Table 4: Best configuration per band family (TWAP, 30–60min). “Combined” multiplies the gamma band by a $\sqrt{R(t)/R(0)}$ time-decay factor; “time-decay” uses that factor alone.

The width trap. The unconstrained winner is the *tightest static band in the grid* ($H_0 = 2$, $L_{\text{std}} = 0.4946$), and the static ladder falls monotonically into that grid edge ($H_0 = 25$: 0.5568; 10: 0.5321; 5: 0.5068; 2: 0.4947). This is the boundary solution of Section 2.6, not evidence that dynamics fail: at matched H_0 the dynamic families look far better than static purely because they are narrower on average. Only the width-matched comparison separates the two effects.

Width-matched comparison. Setting a static band to each dynamic configuration’s mean effective width isolates the timing information, and it splits the two mechanisms cleanly.

Gamma conditioning holds a genuine edge of roughly 2.5–4.0% at moderate widths. The time-decay factor is *negative* at every width: its rationale was that the forced end-of-day flatten is expensive, but the closing-auction discount removes exactly that penalty, so shrinking the band into the close pre-pays

H_0	Mean width	Gamma	Combined	Time-decay
10	4.1	+0.22%	+0.34%	−0.22%
25	10.3	+2.56%	+1.44%	−0.25%
50	20.6	+3.23%	+4.38%	−1.35%
100	41.1	+4.02%	+0.83%	−8.52%
200	82.3	+2.31%	+2.32%	−3.31%

Table 5: Edge over a width-matched static band, % of L_{std} (TWAP 60min). Positive = dynamic better. Mean width in contract equivalents.

execution for nothing. The combined form inherits a diluted positive edge from its gamma component.

Robustness. On three re-seeded tapes (per-day bootstrap re-drawing contract, side and size, with c and the matched widths frozen), the gamma edge is positive on every seed (+1.18%, +0.58%, +0.47%), as is the combined form; time-decay remains mixed.

Band-edge targeting. Replacing the full flatten with the edge target of Equation (4) is the strongest single refinement on ES.

H_0	5	10	25	50	100
Static band	+2.6%	+5.4%	+6.0%	+1.5%	−9.4%
Combined dynamic	+0.3%	+0.8%	+1.8%	+1.7%	+3.7%

Table 6: Improvement from edge targeting (TWAP 60min); positive = better than full flatten.

Parking residual delta at the band edge and carrying it to a cheap auction close is worth up to +6% for moderate static widths, though it backfires for wide static bands, where too much risk is parked under $\lambda_{\text{risk}} = 1$. The dynamic band removes that failure mode — its shrinking width pulls the parked delta inward — and the best overall configuration is the **combined dynamic band with edge targeting** ($H_0 = 5$, $c = 20$: $L_{\text{std}} = 0.4907$, against 0.4946 for the best full-flatten configuration).

4 Results: SPCX Equity

Data. 22 trading days (June 16 – July 17, 2026), 12,092 simulated SPCX option executions (JPM modeled at 1% of option flow), 1,716 five-minute bars. The stock traded \$123.12–\$222.80 (median \$155.27) with ATM implied volatility of 70–126% and a 20-day ADV of $\approx 60.9\text{M}$ shares (\$9.5B/day), from real Nasdaq daily bars. Cost model: measured half-spread \$0.027/share, $\alpha = Y\sigma_{\text{daily}} = 0.0273$, execution-window liquidity $L_k = \text{ADV} \times 5/390$, closing-cross impact haircut, $M_h = 1$.

Only the gamma mechanism is carried over. The time-decay factor

Benchmark	Exec. cost	Risk (RMS/bar)	L_{std}
Zero Hedge	\$154,125	\$226,538	1.015
Full Hedge	\$10,528,317	\$0	1.000

Table 7: SPCX benchmark bounds over 22 days. The selected strategy turns over $\approx 114,000$ shares/day, under 0.2% of ADV. Impact dominates spread ($\approx \$0.69$ vs. $\$0.027$ per share all-in): the binding cost on this book is footprint, not the quoted spread.

is dropped for this book: the Nasdaq closing cross makes the end-of-day flatten cheap, removing the penalty that factor was designed to avoid — the same effect that turned its ES edge negative.

Gamma signal. Median $|\Gamma|$ is 488 shares/\$, with a within-day coefficient of variation of 0.46 and a max/median ratio of only $1.6\times$. Across bars gamma correlates with $|\Delta|$ at $\rho = 0.53$: a good part of the gamma signal is already in delta, and the range over which the band can flex is narrow.

Family	H_0 (sh)	c	Exec. cost	Risk	L_{std}
Static	200,000	—	\$1.74M	\$78,303	0.5110
Gamma	300,000	2×10^{-4}	\$1.74M	\$78,303	0.5110

Table 8: Best configuration per family (VWAP, 30min). Both cut the objective by $\approx 50\%$ against Zero Hedge and $\approx 49\%$ against Full Hedge — the first-order result on this book. Best static and best gamma are *identical* to four decimals: the winning tilt ($c = 2 \times 10^{-4}$) leaves the band static in all but name.

The optimum is a wide, interior band. Because the impact proxy rewards splitting a hedge into ever-smaller children, one must check the optimum is not sitting on the tight edge of the grid. It is not: the static ladder is *U-shaped* with an interior minimum ($H_0 = 5\text{k}$: 0.552; 25k : 0.569; 100k : 0.565; **200k : 0.511**; 300k : 0.712). The best band is *wide* and fires only ≈ 0.4 hedges/day; tighter bands are strictly worse because impact is not negligible. The grid optimum is well-behaved, needing no per-ticket cost or hedge-frequency cap to make it interior.

Width-matched comparison. Width-for-width the gamma band remains mildly risk-efficient across the widths that bind, but the edge does not carry to the optimum.

The operational regime. A hedge-frequency cap does not bind: the grid optimum already fires ≈ 0.4 hedges/day — below even a 5/day cap — so capping changes nothing and static wins or ties at every level.

Because the optimum hedges rarely, there is no scarce hedge budget whose allocation gamma could improve — allocating hedges to high-convexity moments, which Equation (2) optimizes, only helps when hedges are scarce. Across

H_0 (sh)	Mean width	Gamma L	Matched static L	Edge
25,000	16,520	0.5685	0.5793	+1.86%
50,000	33,039	0.5706	0.5730	+0.42%
100,000	66,079	0.5669	0.5806	+2.36%
200,000	132,158	0.5657	0.5755	+1.71%
300,000	198,236	0.5704	0.5441	-4.84%

Table 9: Width-matched gamma vs. static at $c = 10^{-3}$, TWAP 60min. The edge is +0.4% to +2.4% across binding widths and collapses at the widest setting, where the band rarely binds and the tilt merely over-tightens.

Cap/day	Static L	Gamma L	Edge	Winner
5	0.5110	0.5110	0.00%	static
10	0.5110	0.5110	0.00%	static
15	0.5110	0.5110	0.00%	static
20	0.5110	0.5110	0.00%	static
30	0.5110	0.5110	0.00%	static
none	0.5110	0.5110	+0.00%	tie

Table 10: Best static vs. best gamma under a cap on hedge frequency. Every row is identical because the wide-band optimum fires far below the tightest cap.

$\lambda_{\text{risk}} \in [0.25, 4]$ the gamma band likewise only ties or leads the static band by $\leq +0.4\%$.

Band-edge targeting. Mildly positive for moderate widths of either shape, negative only for the widest bands.

H_0 (sh)	10,000	25,000	50,000	100,000	200,000
Static band	+1.04%	+3.50%	+2.04%	+0.52%	-15.02%
Gamma band	+0.86%	+0.80%	+2.33%	+2.12%	-1.17%

Table 11: Improvement from edge targeting (TWAP 60min).

Carrying residual delta to a cheap auction close is most of what edge targeting buys, and on SPCX the closing cross already delivers that to every strategy, so the effect is small and unsystematic — a minor tuning knob, not a driver.

Comparison with ES. The gamma-conditioned band is risk-efficient at a matched width on both books (2.5–4.0% on ES, up to $\approx 2.4\%$ on SPCX), but its operational value differs. On ES a realistic per-ticket cost bites and edge targeting stacks with the band. On SPCX, hedging is cheap relative to the book, the optimum is a wide, rarely-firing band, and neither gamma conditioning nor a hedge-frequency cap changes the ranking. The shared caveat is the frictionless proxy’s lack of a floor on order splitting; SPCX’s short 22-day sample is a further one.